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# Car IDS — Component Purchase List

Intrusion Detection System for Automated Cars (AI + IoT)

**Project: Harshith | Date: 2026-04-27 | Timeline: 3 Days**

## MUST PURCHASE

| # | COMPONENT | QTY | PURPOSE | PRIORITY | BOUGHT |
|---|---|---|---|---|---|
| 1 | **2WD Robot Car Chassis Kit** With wheels, motor mounts, screws, acrylic/metal base | 1 | Physical car frame — motors mount here | P1 | ☐ |
| 2 | **IR LED Transmitter (940nm)** + 100Ω resistor. Standard through-hole IR LED. | 2 | Traffic signal emitter — sends IR pulse on RED | P1 | ☐ |
| 3 | **ESP32 Dev Board** ESP32-WROOM-32 or ESP32-DevKitC. 38-pin preferred. | 1 | Second ESP32 as traffic signal controller | P1 | ☐ |
| 4 | **7.4V LiPo Battery (1000–2000 mAh)** 2S LiPo with JST or XT30 connector + compatible charger | 1 | Main power — drives motors + ESP32 via buck converter | P1 | ☐ |
| 5 | **Rotary Encoder Module (LM393)** Hall effect or optical disk type — motor shaft mount | 2 | Accurate real-time speed measurement per motor | P2 | ☐ |
| 6 | **0.96" OLED Display (SSD1306, I2C)** 128×64 px, I2C address 0x3C, 3.3V compatible | 1 | Shows SAFE / INTRUSION on car — strong demo visual | P2 | ☐ |
| 7 | **Jumper Wire Pack** Male-Male, Male-Female, Female-Female, 20cm each | 1 pack | All wiring between ESP32, sensors, motor driver | P2 | ☐ |
| 8 | **Half-size Breadboard (400 tie points)** Standard 400-point solderless breadboard | 2 | Prototyping connections — one per ESP32 | P2 | ☐ |

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|---|---|---|---|---|---|
| 9 | **DC Barrel Jack / XT30 Connector**<br>Female XT30 or 5.5×2.1mm DC barrel jack panel mount | 1 | Clean battery connection to buck converter input | P3 | ☐ |
| 10 | **Buzzer (Active, 5V)**<br>Active piezo buzzer, 5V, through-hole | 1 | Audible alert when intrusion detected — demo impact | P3 | ☐ |
| 11 | **Red + Green LED (5mm)**<br>2× red, 2× green, through-hole, with 220Ω resistors | 1 pack | Traffic signal visual indicator on signal controller unit | P3 | ☐ |

## ALREADY HAVE — VERIFY BEFORE SHOPPING

| # | COMPONENT | QTY NEEDED | ROLE IN PROJECT | VERIFIED |
|---|---|---|---|---|
| — | **ESP32 Dev Board**<br>Any ESP32 variant | 1 | Main car controller | ✓ |
| — | **IR Receiver Module**<br>TSOP1738 or equivalent 38kHz demodulator | 1 | Detects traffic signal on car | ✓ |
| — | **DC Motors**<br>3V–6V DC geared motors | 2 | Drive wheels | ✓ |
| — | **L298N Motor Driver**<br>Dual H-bridge, handles 2 motors | 1 | PWM motor speed control | ✓ |
| — | **Buck Converter**<br>Adjustable DC-DC step-down (LM2596 or similar) | 1 | Regulate 7.4V → 5V for ESP32 + logic | ✓ |
| — | **HC-SR04 Ultrasonic Sensor**<br>5V, 2cm–400cm range | 1 | Emergency obstacle stop (<15cm) | ✓ |

## SOFTWARE SETUP (NO PURCHASE NEEDED)

| SOFTWARE | INSTALL COMMAND | PURPOSE |
|---|---|---|
| **Mosquitto MQTT Broker** | `brew install mosquitto` | Message broker on laptop |
| **Python 3.11+** | `brew install python` | IDS service + dashboard |
| **Arduino IDE 2.x** | arduino.cc/en/software | Flash ESP32 firmware |
| **MQTT Explorer** | mqtt-explorer.com | Debug MQTT messages visually |

**Arduino Libraries to Install (via Library Manager in Arduino IDE)**

PubSubClient — by Nick O'Leary | NewPing — by Tim Eckel | IRremote — by Armin Joachimsmeyer (v4.x) | Adafruit SSD1306 — by Adafruit (for OLED)

Car IDS Project — Harshith

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